

Answer: A

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$$\sin \theta = \frac{\lambda}{b} \Rightarrow \sin \theta = \frac{600 \times 10^{-9}}{0.1 \times 10^{-3}} \Rightarrow \theta = 0.34378^\circ$$

$$\tan \theta = \frac{x}{3.0} \Rightarrow x = 3.0 \times \tan(0.34378^\circ) \approx 0.018$$

$$\text{width} = 2(0.018) = 0.036 \text{ m}$$

Answer: C

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